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# DrowsyGuard

## Edge-AI Driver Drowsiness Detection on Arduino UNO Q

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Platform: **Arduino UNO Q** – Qualcomm Dragonwing QRB2210 (Linux) + STM32U585 (MCU)

**Qualcomm Future Makers: Hack The Challenge 2026 · Team ML IoT**

# Agenda

**PART 1**
**The Problem**
What we are solving and who suffers from it.

**PART 2**
**The Product**
What DrowsyGuard is, does and how it is built.

**PART 3**
**Technical Readiness**
What already works and what is left.

**PART 4**
**Application Domains**
Where it applies and how far it scales.

**Camera sees the driver** → **Edge AI detects drowsiness** → **Escalating alert & control**

100% on-device | No internet needed | Privacy preserved

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PART 1

# The Problem We Solve

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# Drowsy driving is a silent killer

### Why it matters

- Fatigue and micro-sleep are among the leading causes of severe road accidents worldwide.
- Highest risk for **long-haul truck, coach and ride-hailing drivers**, especially at night.
- A micro-sleep of just **3–4 seconds** at 90 km/h means driving **blind for 100 m**.

### Vietnam context

Insert local statistics on fatigue-related traffic accidents here (source to be cited). The pain is concrete and nationwide.


*A fatigued driver at the wheel, late at night.*

# Who suffers from it

### Long-haul drivers

Night runs, tight delivery windows, hours alone at the wheel. The highest-risk group.

### Coach & bus operators

Dozens of passengers per vehicle – one micro-sleep becomes a mass-casualty event.

### Ride-hailing & private

Long shifts to hit income targets, often after a full working day.

### And the businesses behind them

Fleet and logistics operators carry the cost: vehicle damage, cargo loss, insurance premiums, driver injury and reputational damage after every fatigue incident.

# Existing solutions leave a gap

| Approach | Limitation | Consequence |
|---|---|---|
| Built-in driver monitoring on premium cars | Only on expensive vehicles | Out of reach for most drivers |
| Cloud dashcam analytics | Needs network, adds latency | Fails on rural roads, feels intrusive |
| Wearables (smart bands, caps) | Uncomfortable, easy to forget | Low adoption in practice |

**◎ The unmet need**

A **low-cost add-on device** that runs **fully on-device**, works **offline**, protects **privacy**, and fits **any vehicle**.

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PART 2

# The Product

# DrowsyGuard in one line

## Concept

A small dashboard device that **watches the driver with a camera**, detects drowsiness **on the edge**, and triggers an **escalating alert** before it is too late.

### Perceive

Retrained YOLO detects eyes, yawns and head nods in real time on the Linux core.

### Decide

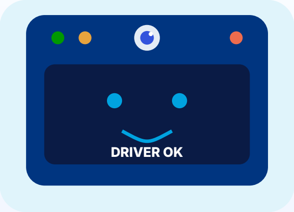
PERCLOS, yawn rate and nod count are fused into a drowsiness level 0–3.

### Act

The MCU escalates the alert: beep, then haptics, then full alarm and SOS.

Low latency | Offline | Private by design | Fits any car

# What the product looks like




Compact unit mounts on the dashboard, camera facing the driver.

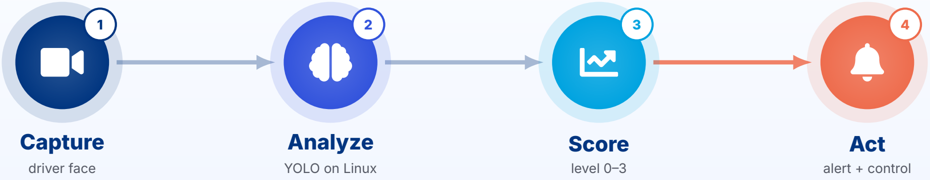
### Simple status interface

- Green: driver alert.
- Amber: early drowsiness.
- Red: danger, full alarm.

### Zero-friction UX

Plug into 12 V power and it just runs. No pairing, no app setup, no driver action required.

# How it works: end to end



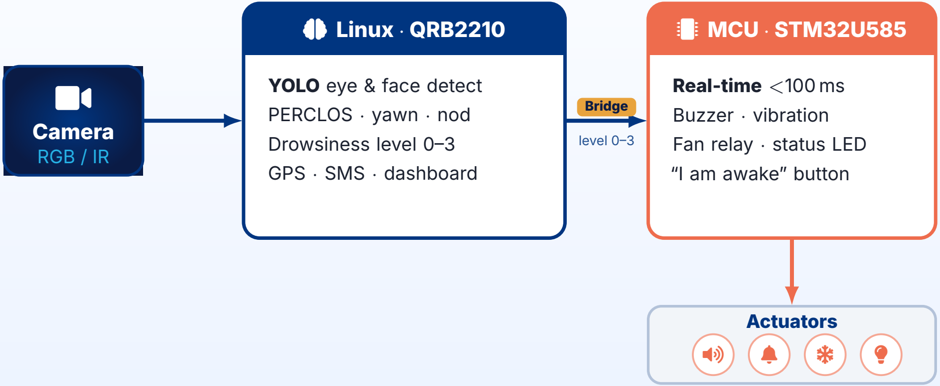

### Everything stays on the device

No frame ever leaves the unit. All four steps run locally, so the system reacts instantly and keeps the driver's face private.

### Reaction budget

Target end-to-end latency **under 200 ms** from eyes closing to the first alert – impossible to guarantee over the cloud.

# System architecture: the dual-brain board




Linux for heavy AI, MCU for instant control – one Arduino UNO Q board does both

# Escalating alert ladder

**L3 Danger**
Hazard, slowdown, SOS

**L2 Drowsy**
Alarm, haptics, fan

**L1 Early**
Beep, amber LED

**L0 Normal**
Silent, green LED

**increasing drowsiness**

## Design principle

Gentle enough not to startle, strong enough to wake. The driver presses **"I am awake"** to reset.

# Control functions

### On the MCU – real time

- Buzzer and voice warning.
- Vibration motor in seat / steering.
- Relay for cool-air fan.
- Three-color status LED.
- “I am awake” acknowledge button.
- GPIO/CAN signal to cut throttle or flash hazards.

### On Linux – connectivity

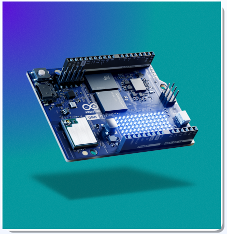
- Log drowsiness events with time + GPS.
- Send SOS via SMS / 4G if unresponsive.
- Fleet dashboard with per-driver safety score.
- Per-driver threshold personalization.

### Fleet value

Logistics companies get a safety tool, not just a gadget.

# Hardware and cost

| Component | Role |
|---|---|
| Arduino UNO Q | Compute (AI + MCU) |
| RGB + IR camera | Day/night capture |
| Buzzer + speaker | Audio alert |
| Vibration motor | Haptic alert |
| Fan + relay | Cool-air wake-up |
| GPS + status LED | Location + UI |



## $ Cost position

A **low-cost add-on**, a fraction of the price of built-in premium driver-monitoring systems. One self-contained unit, powered from the 12 V socket.

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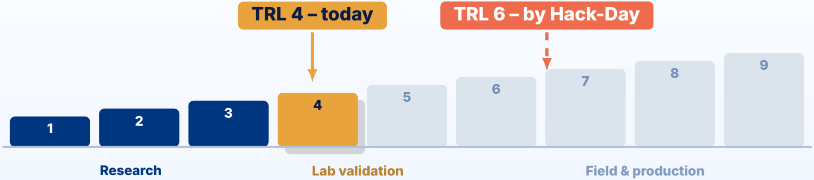
PART 3

# Technical Readiness

# Where we stand: Technology Readiness Level




### TRL 4 today – validated in lab

Detection pipeline proven on public datasets and on Arduino's own on-camera detector; alert logic specified end to end.

### TRL 6 target – demo in a real cabin

With the loaned UNO Q we integrate camera, model and actuators into one unit and demonstrate it in a vehicle environment.

# What already works today

## Proven building blocks

- **On-device detection on UNO Q** – proven by App Lab.
- **Datasets secured** – labelled, YOLO-ready.
- **Toolchain installed** – Arduino CLI, Flasher, AI Hub.
- **Alert logic specified** – PERCLOS + 4-level ladder.

## What the hardware unlocks

Measure real FPS, wire the actuators and close the loop in a cabin.


Arduino App Lab running an on-camera detector on the UNO Q itself.

# AI model: a retrained YOLO detector

## Detection classes

- **Eye open** vs **eye closed**
- **Yawn** (wide open mouth)
- **Head nod / down**

## Why YOLO?

Fast single-shot detector – ideal for real-time inference on an edge device.

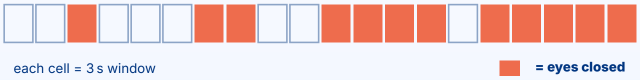
## Day and night

RGB by day, infrared at night – works in a dark cabin.

The model tracks eye and mouth landmarks on every frame.

# From frames to a drowsiness decision

**PERCLOS** = fraction of time the eyes are closed over a rolling 60-second window.




### PERCLOS

Rising closed-eye ratio is the strongest early sign of fatigue.

### Yawn rate

Frequent yawns add to the drowsiness score.

### Head nods

Repeated nodding pushes the level toward danger.

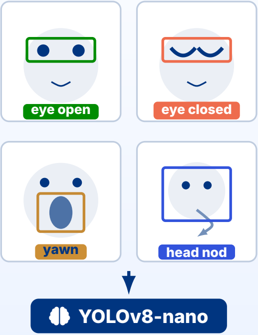
# Training data

| Dataset | Content | Use |
|---|---|---|
| Roboflow drowsiness / fatigue | Drowsy vs awake | Base classes |
| Roboflow eye state | Eyes open / closed | Class balance |
| Car-driver set | In-cabin footage | Fine-tuning |

### Data strategy

Merge public sets, **balance open vs closed**, augment for glasses, masks and low light. Exports directly to YOLOv8 and TFLite.

**Annotated training frames**




*Labelled eye/mouth frames teach the detector each state.*

# Optimized to run on the edge

### Model optimization

- YOLO **nano** backbone – small and fast.
- **INT8 quantization** for CPU/GPU inference.
- Runs on QRB2210 (no large NPU needed).

### State the numbers

Report a concrete figure, e.g. *"nano INT8 at ≈ N FPS on Cortex-A53"*. This proves it truly runs, not just in theory.


**Tool: Qualcomm AI Hub for profiling**

# Risks and how we handle them

| Risk | Mitigation |
|---|---|
| Dark cabin at night | Infrared camera + IR illumination |
| Driver wears glasses or a mask | Augment training data for both cases |
| Vibration, heat and dust in the cabin | Rugged mounting, automotive-grade parts |
| False alarms annoy the driver | Gradual ladder + “I am awake” acknowledge |
| Inference too slow on device | Nano backbone, INT8, resolution tuning |

# Roadmap – aligned to the competition

we are here

| 10–30 Jul | 1–7 Aug | 7–15 Aug | 16 Aug | 20 Aug |
|---|---|---|---|---|
| **Application** | **Top 10** | **Development** | **Hack-Day** | **Award** |
| Submit our proposal | Shortlist selection | Build & flash on UNO Q | Live pitch & demo | Award ceremony |

🏁 **Our goal: a working UNO Q prototype demoed live at Hack-Day, 16 Aug 2026**

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PART 4

# Application Domains

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# Primary domain: road safety

### Commercial fleets first

- Logistics and long-haul trucking.
- Coach and inter-province bus operators.
- Ride-hailing and taxi fleets.

### Why fleets are the entry point

They already pay for safety and telematics, buy in volume, and can measure the return directly in avoided incidents.


*One device per cab, retrofitted to any vehicle.*

# The same technology travels far

### Industry

Crane, forklift and heavy-machinery operators on long shifts.

### Rail & marine

Train drivers and ship watchkeepers on monotonous night duty.

### Mining & construction

Haul-truck drivers on remote sites with no network coverage.

### Safety-critical roles

Control-room staff and security guards on night watch.

### One core, many verticals

The same pipeline – **camera** → **on-device vision** → **fatigue score** → **escalating alert** – applies anywhere a human must stay alert. Only the mounting and the action at level 3 change.

# Impact and how it scales

## Impact

- Fewer fatigue accidents – lives saved.
- Lower cost and insurance risk for carriers.
- Affordable safety for everyday vehicles.

## Scale-up path

- Distraction alert (phone use at the wheel).
- Fleet-wide safety scoring.
- Insurance and telematics integration.

**Save lives** **Cut costs** **Ready to grow into a platform**

# Demo scenario

### Live demonstration

1. A volunteer sits in front of the device as “driver”.
2. They slowly close their eyes / simulate a yawn.
3. L1: gentle beep and amber LED.
4. L2: loud alarm plus vibration and fan.
5. Press “I am awake” to reset to green.

### Proof plan

We show the escalation live on the board and log the event on the dashboard.

### Backup

A short recorded demo video is prepared in case of hardware issues on stage.

### Media placeholder

Drop a real prototype photo or screenshot here before the final round.

# The team: ML IoT

**Nguyen Hoang Trieu**
Team Lead & Embedded

**Tang Phon Thinh**
AI & Computer Vision

**Van Dac Phong Truc**
Software & Integration

**Vo Phuc Thinh**
Connectivity & Cloud

**Team synergy**

End-to-end ownership: **camera** → **Edge AI** → **decision** → **MCU control** → **fleet dashboard**, built by one focused team.

# In summary

### Problem

Fatigue at the wheel kills, and affordable protection does not exist for ordinary vehicles.

### Product

A dashboard device that sees drowsiness on-device and escalates the alert in under 200 ms.

### Readiness

TRL 4 today, TRL 6 by Hack-Day: proven blocks, datasets and toolchain already in place.

### Domains

Fleets first, then industry, rail, marine and mining – anywhere alertness is safety-critical.

### What we ask for

The loaned **Arduino UNO Q**, Qualcomm SDK access and engineer mentorship – to turn a validated pipeline into a working in-cabin prototype by 16 August.

# Thank You

## Acknowledgements

We thank the Qualcomm Future Makers program and the organizers of Hack The Challenge 2026 for the opportunity, hardware and mentorship.

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